Project Requirement Document (PRD)

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Project Name: ScholarFlow AI

Version: 1.0

1. Project Overview

- **Problem Statement:** Academic and R&D researchers face a highly fragmented, manual workflow for literature reviews. Currently, researchers must manually search multiple databases, navigate paywalls, read abstracts for relevance, and synthesize findings.
- **Target Domain:** Automating the literature review and R&D methodology discovery process.
- **Proposed Solution Summary:** ScholarFlow AI, a cloud-based engine that moves the user from "manual laborer" to "curator." It automates the discovery, filtering, and synthesis of research papers.

2. Background & Business Objective

- **Background of the Problem:** Traditionally, researchers follow a manual linear pathway: manual searching, downloading, filtering, and synthesis. This process is time-consuming, prone to human error, and limits the volume of research one can effectively analyze.
- **Importance of Solving This Issue:** Accelerates the time to synthesize research from days to minutes, allowing researchers to iterate on literature reviews rapidly. It lowers the barrier to entry for junior researchers who may struggle with managing large volumes of academic information.
- **Strategic Fit:** Aligns with AI-Native research workflows. By leveraging cloud-based APIs (OpenAlex/ArXiv), it creates a seamless integration between raw data and synthesized markdown reports.

3. Product Purpose

- **Main Goal of the System:** To provide an AI-enabled research partner that automates the literature review lifecycle.
- **Intended Users (Target Audience):** * Academic Researchers & University Students.
    - R&D Engineers & Data Scientists.
    - Corporate Analysts.

4. System Functionalities

- **Description:** An AI-powered research engine that translates natural language queries into automated research synthesis workflows using a "Two-Step Approval" mechanism.

- **Key Functionalities:**
    - **Automated Discovery:** Translates natural language into Boolean logic for OpenAlex/ArXiv.
    - **Manual AI Gatekeeper (Phase 1):** Interactive dashboard to review relevance scores before deep analysis.
    - **Context-Aware PDF Chunking (Phase 2):** Downloads and processes PDFs, splitting them into token-safe chunks for deep analysis.
    - **Real-Time Streaming:** Uses Server-Sent Events (SSE) for progress visualization.
    - **Dynamic Caching:** Implements chunk_cache in PostgreSQL to optimize costs.

5. AI Model & Prompt Design

- **Model Selection:** Powered by **Z AI GLM** (ilmu-glm-5.1). Chosen for its high-performance synthesis capabilities and context window efficiency.
- **Prompting Strategy:** * *Discovery Phase:* Zero-Shot Prompting to score relevance (0-100) and generate 3-sentence insights.
    - *Deep Analysis Phase:* Multi-step agentic chunking to manage documents exceeding token limits.
- **Context & Input Handling:** The system manages large PDFs by splitting content into token-safe chunks before processing. Chunks exceeding the model's window are dynamically truncated or skipped based on semantic significance.
- **Fallback & Failure Behavior:** If the model returns unusable responses, the system triggers a retry with a refined, high-temperature prompt. A human escalation path exists in the UI to allow users to manually re-trigger specific sections.

6. User Stories & Use Cases

- **User Story:** "As an academic researcher, I want to search for topics using my own words, so that the AI automatically filters irrelevant papers and synthesizes the core findings into a Markdown report for me."
- **Use Cases (Main Interactions):**
    - **Query to Insight:** User inputs query → System generates gallery of relevant papers with 0-100 scores.
    - **Deep Synthesis:** User selects paper → System extracts, chunks, and summarizes content into a comprehensive report.

7. Features Included (Scope Definition)

- OpenAlex/ArXiv API integrations.
- Two-Step AI workflow (Discovery & Deep Synthesis).
- SSE-based real-time progress monitoring.
- Markdown synthesis generation.

8. Features Not Included (Scope Control)

- User authentication/login systems.
- Document exports to proprietary formats (DOCX/PDF).
- Integration with paid proprietary databases (e.g., IEEE Xplore).

9. Assumptions & Constraints

- **LLM Cost Constraint:** Estimated at [X] tokens per average user session. To manage costs, the chunk_cache prevents re-processing identical PDF segments.
- **Technical Constraints:** Synthesis output is limited to Markdown format. Advanced citation formatting (e.g., BibTeX auto-generation) is not yet supported.
- **Performance Constraints:** Processing time is dependent on PDF length and model latency; strict chunking limits are enforced.

10. Risks & Questions Throughout Development

- **Hallucination:** How do we ensure the AI accurately cites the paper rather than synthesizing potentially incorrect information?
- **Accessibility:** How does the system handle papers that are not open-access if ArXiv fallback fails?
- **Relevance:** How do we optimize the Boolean logic translation to ensure "high-quality" papers are prioritized in the initial search?